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Executive Summary

This report assesses the economic costs and benefits of decisions made by Australian governments to invest in eradication of the August 2007 equine influenza (EI) outbreak and in ongoing biosecurity arrangements to maintain EI freedom.

Ongoing biosecurity controls have generated benefits of \$879 million in avoided costs for the horse industry and horse community, far exceeding the \$14 million in costs for prevention and monitoring. The net benefits of ongoing biosecurity were therefore \$865 million, and the investment had a benefit-cost ratio of 63 to 1. Costs for eradication of the outbreak were \$571 million, including compensation payments to offset disruption costs to the horse industry/community. Comparing the sizeable net benefits from maintaining EI freedom (\$865 million) against the eradication costs (\$571 million) suggests the overall net benefits of government biosecurity investments could be at least \$294 million with the investment producing a benefit-cost ratio of 1.5 to 1. These findings demonstrate that eradicating the outbreak was a good decision by Australian governments and that continued investments in prevention and monitoring are warranted.

The report comprises two components: a literature review of previous studies to estimate the costs and benefits associated with eradication of the 2007 outbreak and a modelling exercise to estimate the economic value of maintaining EI freedom.

2007 equine influenza outbreak in Australia

Known also as "horse flu" and "A1 influenza", equine influenza (EI) is an acute, highly contagious, viral respiratory disease in horses and other equine species. It spreads rapidly and is expensive to eradicate, can disrupt equine events and cause significant economic disruption to horse related industries. New strains of the disease mean that vaccines need to be updated regularly. EI is unlikely to cause deaths in healthy adult horses and humans are not susceptible to the virus. The disease is endemic throughout most of the world, except for Australia, New Zealand and Iceland (OIE 2020).

In August 2007, the EI virus was detected in Australia, and within a matter of months had infected around 69,000 horses on 9,600 premises in New South Wales and Queensland. To eradicate the virus, a nationwide eradication program to control its spread and regain Australia's disease-free status was implemented.

Preserving Australia's disease-free status has required significant resources

Maintaining EI freedom has not been trivial. The 2007 outbreak posed a serious risk to the Australian horse population and the interests of the horse industry and horse community. Around 7% of the horse population was infected. To manage this threat, Australia initiated a comprehensive biosecurity response.

The literature review established that the \$571 million eradication costs incurred by Australian governments included compensation payments of \$370 million to offset disruption costs to the horse industry and horse community during the eradication period. The Commonwealth Government and jurisdictions incurred \$435 million and \$136 million, respectively.

Disruption costs reportedly range from \$30 million to \$1.5 billion. However, none of these estimates represent a serious attempt at quantifying the true economic costs of the disruption. For example, the larger estimates include losses to individual businesses which are more than likely to be (almost) fully offset by gains to other businesses. This arises from consumers spending money on non-equine-related activities during the disruption. The true economic costs arise from the dead weight losses of productive resources not being utilised and the (marginal) difference in consumer benefit from the less-preferred spending pattern. While compensation payments are transfer payments, they can also be treated as a proxy for real costs incurred by the horse industry and horse community.

In addition to economic costs, movement restrictions and other quarantine measures also caused some social costs in the form of impacts on the psychological wellbeing of horse owners and people directly employed in the industry. A review of the literature found evidence of social and emotional impacts felt across the horse community, particularly those in the red and amber EI control zones in New South Wales and Queensland.

Since Australia's declaration of EI freedom, Commonwealth, state and territory governments have worked together to implement a number of biosecurity measures to prevent EI's re-entry into Australia. The modelling exercise estimated that ongoing prevention and monitoring post eradication have cost \$14 million over the last 12 years. This represents \$1.87 million per year of which \$1.71 million per year

(92%) was cost recovered by the Commonwealth Government from horse importers for quarantining horses pre-border (before departing the exporting country) and at-border (in Australia upon arrival). The remaining \$160,000 per year was incurred for post-border monitoring, generally undertaken by the jurisdictions.

EI biosecurity arrangements have been worthwhile and effective

Since the initial outbreak, Australia has remained free from EI. To estimate the overall net benefits of Australian government biosecurity arrangements since 2007, the benefits (or avoided costs) of ongoing biosecurity arrangements were compared against the costs associated with eradication (including compensation payments), and prevention and monitoring activities.

The \$879 million benefits of ongoing biosecurity control reported above were estimated by comparing a base case and counterfactual scenario for the post eradication period. In the base case, Australia remained EI-free due to biosecurity arrangements that prevented the entry of EI over the past 12 years. The counterfactual simulated what could have happened during this time in the absence of those arrangements. The findings demonstrate that government biosecurity investments over the 12 years since declaration of EI freedom have been both worthwhile and cost effective. Additionally, every year that Australia is free of EI after these 12 years adds to these returns.

Eradication costs after a significant outbreak exceed the cost of maintaining an effective prevention system for hundreds of years

Effective prevention and monitoring measures are relatively cheap to maintain as opposed to eradication costs if an EI outbreak were to occur. For example, the 2007 outbreak cost \$571 million to eradicate, compared to just \$14 million spent over the following 12 years, or \$1.87 million per year in ongoing prevention and monitoring measures.

This result shows the crucial importance of funding a strong prevention and monitoring regime to stop EI from re-entering Australia. Eradication costs in response to a future similar outbreak would be equivalent to funding hundreds of years of effective prevention and monitoring activities.

A positive net present value for maintaining EI freedom

There are a number of uncertainties in the modelling. These include the underlying size of the horse population, the probability of EI entry into Australia along with vaccination rates and efficacy. Sensitivity tests were undertaken on these aspects and the findings highlighted above were found to be robust to a wide range of plausible scenarios. There were also uncertainties in the literature review regarding the economic losses from disruption to the horse industry and horse community.

Irrespective of the probability of EI entry, the discount rate and vaccination rate and efficacy applied or variations in modelled horse population estimates, all results show a finding of positive net benefits for the decisions made by Australian governments to invest in keeping Australia free from EI.

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